KisanShakti: An Integrated Smart Farming and Hyperlocal Commerce Platform for Enhancing Farmer Profitability in India

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***Abstract*—Nearly half of India's workforce is engaged in agriculture however, farmers frequently encounter challenges related to unpredictable income due to their continued reliance on traditional, heuristic based farming practices. Contemporary agricultural applications tend to address isolated aspects of these challenges, leaving smallholder farmers vulnerable to abrupt climatic changes, crop diseases, and inequitable supply chains, where intermediaries capture 60-70% of the crop's final value. In response to these issues, our research presents "KisanShakti," an integrated mobile and IoT platform designed to equip farmers with advanced farming tools and facilitate direct access to local buyers. The app's architecture uses Artificial Intelligence to diagnose crop diseases directly from a phone camera, even without an internet connection, and connects with IoT sensors to monitor field conditions in real time. The system uses location-matching algorithms to connect farmers with nearby retail outlets while analysing weather and market trends to offer tailored crop and livestock management recommendations. KisanShakti aims to enable proactive planning, reduce agricultural costs, preserve crops through early disease detection, and increase profits by cutting intermediary fees. It transforms traditional smallholders into resilient entrepreneurs, offering a framework for rural prosperity and sustainable agriculture.**

Keywords—Digital Agriculture, Precision Farming, Hyperlocal Commerce, Smallholder Profitability, IoT Integration.

I. INTRODUCTION:

Agriculture faces challenges from climate change, resource scarcity, population growth, and labour shortages, driving the sector toward digital and smart farming to boost productivity and sustainability[1] [2].

Smallholder farmers, crucial to global food security but hindered by low productivity, limited credit, and increasing climate risks, are a primary focus of digital agriculture initiatives [3] [4].

Digital technologies like mobile platforms, IoT sensors, AI, and online marketplaces are transforming how farmers access information, manage resources, and connect to markets [5] [1] [6] [7] [8]. Precision agriculture and smart sensing enable real-time monitoring of soil, crops, and the environment, allowing farmers to make data driven decisions on irrigation, fertilisation, and pest and disease control [1] [9] [6] [8] [10] [11] [12]. These tools can reduce input use, lower costs, and boost yields while promoting climate smart and sustainable practices [1] [9] [13] [8] [10] [14] [12].

Digital platforms and ICT-based extension services tackle long-standing issues like fragmented markets, information asymmetry, and weak advisory systems by enhancing market access, transparency, and expert advice dissemination [5] [15] [17] [7] [18] [19] [20]. However, digital divides in connectivity, affordability, device access, and skills, often defined by gender and social-economic factors, hinder adoption, particularly in rural and smallholder settings [5] [3] [4] [6] [16] [7] [2]. Consequently, policy frameworks, public-private partnerships, and user-centric design are essential to ensure an inclusive digital transformation that promotes food security, rural livelihoods, and the Sustainable Development Goals [5] [3] [15] [4] [7] [18] [19] [2] [20]. Digitalisation provides powerful tools to enhance productivity, resilience, and sustainability in agriculture, especially for smallholders, but its potential relies on overcoming infrastructure, literacy, and equity barriers through supportive policy, inclusive design, and coordinated governance.

II. LITERATURE REVIEW:

The digital transformation of agriculture is increasingly positioned as a pivotal strategy for mitigating climate risks and alleviating rural poverty. Before examining the evidence, it is important to define the core concepts framing this review. **Precision Agriculture (PA)** refers to the use of technology to monitor and manage spatial and temporal variability in fields with the goal of improving crop performance and environmental quality [6][22]. The **Internet of Things (IoT)** describes networks of physical sensors and devices that collect and transmit real-time field data such as soil moisture, temperature, and pest activity [6][21]. **Artificial Intelligence (AI)** in this context encompasses machine learning and computer vision algorithms applied to agricultural data for predictive analytics and automated decision-making [22][8]. A **smallholder farmer** is defined here as a producer cultivating fewer than two hectares, typically operating with limited capital, connectivity, and institutional support [4][26]. Finally, **Integrated Farming Systems (IFS)** refers to mixed crop-livestock operations that are common among rural smallholders and provide economic diversification and resilience [25][30].

This review is organised thematically into three sections: (1) Precision and Smart Farming Technologies, (2) Digital Market Linkages and Hyperlocal Commerce, and (3) The Digital Divide in Smallholder Ecosystems. This thematic structure was chosen because the literature clusters naturally around these three distinct but interrelated problem areas rather than around a single chronological or methodological axis.

Regarding scope, this review **includes** peer-reviewed literature published between 2023 and 2025 addressing PA, IoT/AI integration, digital market platforms, and barriers to adoption among smallholder farmers in low-to-middle-income contexts. It **excludes** literature focused on large-scale commercial agriculture, literature predating 2023 (except foundational references where no recent equivalent exists), grey literature, policy documents, and studies focused exclusively on high-income agricultural economies where infrastructure and capital constraints are not comparable.

A. Precision and Smart Farming Technologies

The most consistently supported finding in recent literature is that IoT-AI integration yields measurable agronomic benefits. Multiple field-based studies confirm that combining IoT sensors with AI-driven analytics enables real-time monitoring of soil moisture, crop health, pest outbreaks, and environmental conditions, producing input savings of up to 45% in water use and yield increases of up to 22% [6][21][22][29]. Computer vision approaches particularly Convolutional Neural Networks (CNNs) have achieved pest and disease detection accuracies exceeding 90% [6][22]. These are significant results that demonstrate the technological maturity of precision tools under controlled or well-resourced conditions.

However, a critical weakness running through this body of work is its narrow scope of applicability. The strength of studies such as Mansoor et al. [6] and Kaur et al. [21] lies in their empirical rigor and quantified outcomes. Their limitation, however, is that most implementations are siloed designed for single crops, single production stages, or environments with reliable cloud connectivity conditions that do not reflect the reality of most smallholder farms [4][25]. Miller et al. [8] partially address this by examining edge computing solutions that enable offline or local data processing, representing a meaningful methodological step toward context-appropriate design, though deployment at scale remains unproven. Overall, the precision farming literature demonstrates strong proof-of-concept but has not yet resolved the gap between laboratory or pilot performance and broad smallholder applicability.

B. Digital Market Linkages and Hyperlocal Commerce

A second theme in recent literature concerns whether digital platforms can improve market outcomes for smallholders. The evidence here is moderate and more contested. Yuan and Sun [33] and Sunarjo et al. [31] find that e-commerce and mobile advisory platforms increase price transparency and reduce dependence on traditional intermediaries, improving net returns in some contexts. These findings are plausible and consistent, but their critical weakness is that most studies evaluate platforms built for bulk trading or large aggregators a design mismatch for the hyperlocal, small-volume producer. Yuan and Sun [3] explicitly acknowledge that frameworks enabling individual smallholders to connect directly with nearby consumers remain rare and underdeveloped, which limits the generalizability of positive market outcomes to the farmers who need them most.

Blockchain-based traceability has attracted research attention as a means of guaranteeing supply chain transparency, but Miller et al. [8] note that pilots face persistent cost and interoperability barriers that have prevented meaningful scaling. This is an area where the literature is notably optimistic in framing but thin in demonstrated outcomes, a gap that warrants methodological caution when interpreting claims about blockchain's transformative potential.

C. The Digital Divide in Smallholder Ecosystems

The third and arguably most critical theme is the structural barrier landscape that constrains the adoption of the technologies discussed in the two preceding sections. This literature is the most sociologically grounded and provides the most important corrective to technologically optimistic accounts. Gumbi et al. [4], Mushi et al. [25], Abdulai et al. [26], and Mehrabi et al. [28] converge on a consistent set of barriers: device and service unaffordability, broadband coverage below 40% for farms under one hectare, low digital literacy rates, gender disparities in technology access, and fragmented or absent training programs.

A particular strength of Abdulai et al. [26] is its use of lived experience, field-based methodology in rural northern Ghana, which surfaces the gap between policy-level digital inclusion narratives and on-the-ground realities, a dimension that survey-based or secondary studies miss. The critical gap identified most sharply by Mushi et al. [8] is that virtually all digital tools are designed exclusively for crop production, ignoring the integrated crop-livestock systems that constitute a critical income-resilience strategy for the majority of smallholders [30]. This systemic design blindspot means that even where adoption occurs, it addresses only part of a farmer's livelihood system. Mushi et al. [25] further identify "app fatigue" the cognitive and practical burden of managing multiple disconnected applications as a meaningful deterrent to sustained use, though this claim rests on a narrower evidence base than the infrastructure barriers.

B. Plan of Action

This review has identified three interconnected dynamics shaping the digital transformation of smallholder agriculture. First, precision and smart farming technologies demonstrate strong technical performance but remain siloed, connectivity-dependent, and decoupled from the mixed farming realities of smallholders [6][21][4][25]. Second, digital market platforms show promise in improving price transparency but are structurally misaligned with the scale and geography of hyperlocal, small-volume producers [3][31]. Third, and most foundationally, persistent barriers of cost, infrastructure, literacy, gender, and fragmented design constrain equitable adoption across all technology categories [8][25][26][28].These converging gaps form the direct motivation for the development of **KisanShakti** a unified, offline-capable agritech platform designed specifically for mixed crop-livestock smallholders operating in low-connectivity environments. KisanShakti integrates IoT-based field monitoring, AI-driven pest and disease detection, a livestock management module, and a geolocation-enabled hyperlocal commerce interface all accessible via a single low-bandwidth application. This architecture responds directly to the fragmentation problem identified by Mushi et al. [25], the connectivity constraints highlighted by Miller et al. [8], the market access gap documented by Yuan and Sun [3], and the integrated farming system neglect noted by Hellin et al. [30]. Participatory design methods are embedded throughout development to ensure the tool reflects actual end-user contexts, with stratified inclusion of women farmers to address the gender disparities documented by Mehrabi et al. [28]. KisanShakti therefore represents a deliberate move beyond the isolated technological fixes that currently dominate the field toward a cohesive, inclusive ecosystem built for the farmers least served by the digital transformation literature to date.

III. PROPOSED SYSTEM ARCHITECTURE:

The KisanShakti platform is a digital system with several layers. It is in the early testing and development stages. Currently, the focus is on the Application Layer, which is the part that users see and interact with. This part was fully developed to test how users interacted with it. The backend, which includes AI and database work, is still in the planning phase. It is being tested with fake data on the front end.

A. Precision and Smart Farming Technologies

The frontend prototype ensures that the app works well for farmers with limited resources. It checks how users interact with it before the full system is set up.

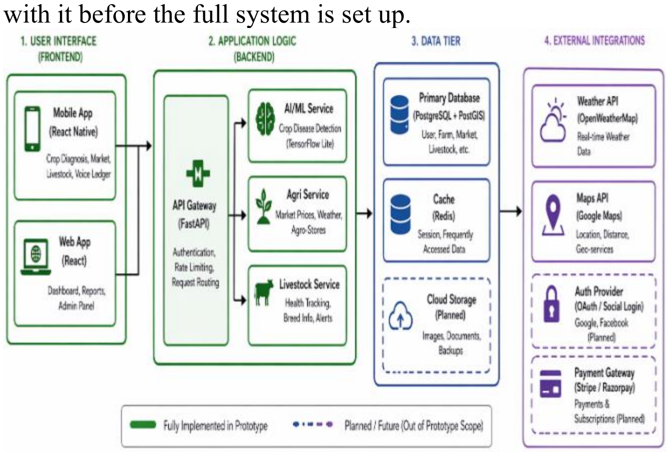

Fig.1. KisanShakti System Architecture and Prototype Scope Overview.

B. Frontend User Experience and Personalised Dashboard

The app starts with a personalised "Home" dashboard. It is designed for people with different levels of digital skills. The interface supports local languages, like Kannada, with greetings such as "Namaskara." The prototype shows real-time weather updates, like "Kolar: Partly Cloudy, Rain Tomorrow." It also displays market price trends for crops like Tomato, Onion, and Ragi using simple graphs.

A key feature is the "Smart Suggestions" engine. It uses the user's profile, like land size and livestock, to give advice. For example, it might suggest "Grow Maize in Plot B this season" if it predicts a 34% higher profit. It also sends alerts like "Vaccinate your cow Lakshmi," helping farmers manage better.

C. Proposed AI-Based Crop Disease Detection

To tackle the shortage of expert agricultural extension services, KisanShakti suggest implementing an offline-capable Convolutional Neural Network(CNN) using TensorFlow Lite.

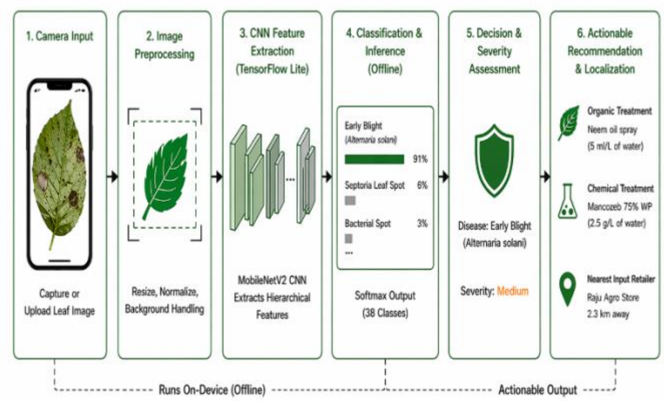

Fig.2. AI-based crop disease detection workflow.

Proposed AI Workflow: In the frontend prototype, this process is verified through a specific "Disease" module. Users can either capture or upload an image of a leaf showing symptoms. The user interface mimics the CNN output by displaying a comprehensive diagnostic card. For instance, it accurately detects "Early Blight (Alternaria solani)" with a simulated confidence level of 91% and a "Medium" severity rating. Importantly, the system offers practical, localised solutions: it suggests both Organic (e.g., Neem oil spray) and Chemical (e.g., Mancozeb 75% WP) treatment options, while using geolocation to recommend the nearest input supplier (e.g., "Raju Agro Store, 2.3 km away").

D. Proposed Hyperlocal Commerce Engine

The "Market" module revolutionises the conventional supply chain dominated by intermediaries by enabling direct sales to retailers and consumers. The frontend prototype includes a simplified listing interface where farmers can enter details about their crops, such as quantity and anticipated pricing, along with uploading a photo of the produce.

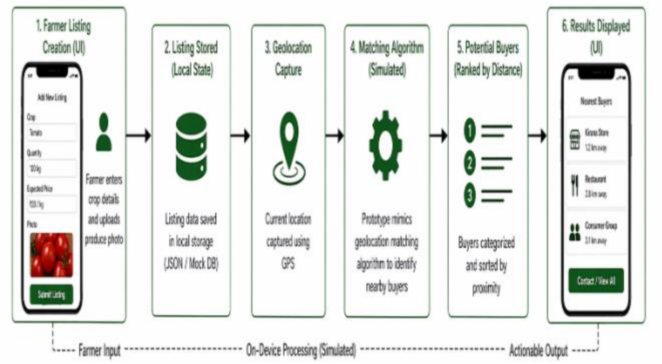

Fig.3. Proposed hyperlocal commerce engine for direct market access and buyer matching.

E. Proposed Market Matching Data Flow

When the "Find Nearest Buyers" action is initiated, the prototype mimics a geolocation matching algorithm. It provides a categorised list of potential buyers sorted by distance (e.g., Kirana Store at 1.2 km, Restaurant at 2.8 km, Consumer Group at 3.1 km). This process confirms the UI flow necessary to bypass APMC mandis and secure higher profit margins for small-scale producers.

Integrated Livestock and IoT Monitoring (My Farm) The "My Farm" module effectively bridges the gap in literature concerning Integrated Farming Systems (IFS) by merging crop infrastructure with livestock management. This prototype serves as an initial point for "IoT Field Monitor" data, simulating sensor connectivity. Below this, the user interface displays digital profiles of livestock. The system monitors individual animal data, including breed (such as HF Cross, Malnad Gidda), daily production outputs (like "Milk today: 8L"), and health conditions. Colour-coded visual indicators (for instance, green for "Healthy" and yellow for "Vaccination Due") offer immediate, easy-to-understand health summaries, demonstrating the concept's capability to manage mixed-farm economic resources in a single platform.

F. Farm Business Intelligence (Profit Tracking)

The “Profit” model functions as a digital ledger and analytics dashboard, treating the smallholder farm as a quantifiable business. The prototype showcases a "Farm Business Overview" that consolidates total Income, Expenses, and Net Profit. The user interface effectively displays interactive bar charts to illustrate "Monthly Profit Trends," aiding farmers in comprehending their cash flow patterns. Additionally, it includes a "Crop-wise Breakdown" that determines the precise Return on Investment (ROI) for each activity (e.g., comparing Tomato income with its specific expenses to achieve a 154% ROI, along with secondary income sources like Milk Sales). This supports the architectural aim of offering farmers clear unit economics.

IV. METHODOLOGY AND PROTOTYPE IMPLEMENTATION:

A. Software Stack & Prototyping Approach

The frontend prototype is crafted using React Native, enabling smooth, native-like deployment on both Android and iOS platforms. The development process adhered to an iterative design methodology, initially concentrating on wireframing, component design, and validating the UI flow to address the "app fatigue" noted in the literature.

B. Simulation of Backend Services

Rather than utilising a live database, the application currently employs structured local data stores, such as local JSON arrays and context hooks, to mimic the backend API responses that will eventually be provided by Python/FastAPI. This approach separates frontend testing from backend dependencies, speeding up the UX validation phase and facilitating offline demonstrations in rural areas.

C. Future Backend Integration

After the frontend interaction models have been thoroughly validated through user focus groups, the subsequent methodological step will be to link the Application Layer with the live Processing Layer. In the upcoming development stages, plans include training TensorFlow Lite models using open-source datasets on plant diseases, deploying PostgreSQL databases for real-time market connections, and incorporating live IoT sensor nodes.

V. EXPECTED OUTCOMES AND IMPACT ANALYSIS:

A. Agronomic Impact

The implementation of the suggested AI diagnostics and IoT monitoring is anticipated to greatly decrease crop losses and enhance fertiliser efficiency by enabling early and accurate interventions.

B. Economic Impact

The hyperlocal commerce engine is projected to significantly boost farmers' net profits by removing the 60-70% margins that are typically taken by supply chain intermediaries.

VI. CONCLUSIONS:

The KisanShakti frontend prototype effectively creates a seamless, user-focused interface that connects intricate agritech solutions with the everyday experiences of smallholder farmers. By providing an integrated dashboard for managing crops, livestock, and markets, the platform directly tackles the issues of fragmentation and app fatigue that currently impede the adoption of digital agriculture. The immediate future scope of this research includes actively developing the proposed backend architecture, training the CNN models, and conducting live field trials to measure the system's impact on rural profitability.

Supplementary materials for KisanShakti, including the project manual, README, execution screenshots, and related reference documents, are available online[34]**.**

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